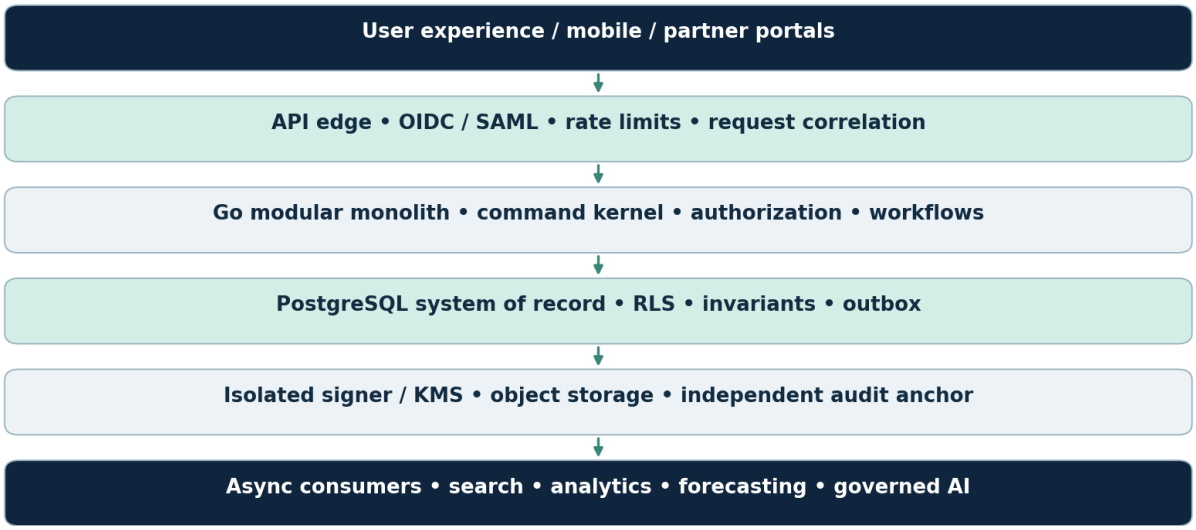
SERK / ENTERPRISE SYSTEMS

CONSTRUCTION INTELLIGENCE OS



Executive determination

Product thesis: a single controlled digital thread from opportunity and BOQ through coordinated project work, resource commitments, commercial entitlements, certified progress, financial accounting and predictive portfolio decisions. The system must remain usable by a small contractor while enforcing independently verifiable control boundaries at enterprise scale.

Release posture: NO-GO. Passing unit tests or attractive architectural diagrams are not production authorization. Independent penetration testing, tenant isolation, accounting reconciliation, field UAT, recovery drills, statutory review and accountable sign-offs are required.

Reading map

Part I: product and architecture (Sections 1–9). Part II: domain detail (Sections 10–17). Part III: acceptance, delivery, evidence and decisions (Sections 18–23). Appendices: complete baseline requirement register, v2 proposal register, standards and source references.

1. Product scope and boundaries

The product shall serve contractors, construction groups and joint ventures working across buildings, roads, water, industrial, MEP and related project types through configuration rather than divergent finance kernels.

In scope	Explicit boundary / out of scope
Enterprise portfolio, programs, contracts, project controls, procurement, stock, plant, workforce, finance, site and compliance	Do not imply proprietary CAD authoring or replace licensed engineering design applications.
BIM/document federation, IFC/BCF exchange, approved revisions and handover	External CDE/modeling integration is permitted; publication control stays with authorized humans.
Contract financial records, bank instructions and statement reconciliation	Bank settlement belongs to financial institutions; no inference that internal payment status means paid.
AI-assisted searches, predictions, risk explanations and drafting	AI has no autonomous authority for posting, contractual issuance, safety sign-off, bank changes or privilege elevation.

1.1 Non-negotiable user outcomes

- A foreman can identify ready and blocked work in under a few interactions, capture evidence without connectivity and safely synchronize later.
- A QS can trace BOQ scope, approved changes, measured quantities, valuation, retention and notices without duplicate data entry.
- A controller can reconcile project commitments, accrued and actual costs, invoices, payment records and general ledger deterministically.

- A PMO can distinguish available from already-reserved crews, plant, materials and funding across projects.
- An auditor can reconstruct who authorized which exact document revision and whether any postings violated policy.

1.2 Multi-tenancy hierarchy

Tenant → legal entity / JV accounting arrangement → portfolio → program → project → contract → WBS / BOQ / work package → activities / resources / approved observations. Portfolio and program are coordination views, not shortcuts around entity-level accounting authority.

2. Source reconciliation and supersession policy

The following v1 invariants are the controlling baseline. v2 risk-tiered approvals cannot silently weaken them.

Invariant	Controlling design and reconciliation
R1 | Ledger truth	Balanced, immutable posted journals with exact money, valid open periods and permitted reversals only.
R2 | Separation of duties	Maker/checker is enforced for all controlled decisions specified by v1; any tiered exception needs a separately approved replacement requirement.
R3 | Human decision authority	Only authenticated humans make restricted decisions; services, automations and AI may propose or prepare, not independently authorize.
R4 | Isolation	Tenant/entity/project constraints enforced at APIs, PostgreSQL, exports, search, objects, integration events and AI retrieval.
R5 | Single business effect	Idempotency within authoritative transaction + effect-level deduplication; delivery itself is at-least-once, not exactly-once messaging.
R6 | Offline staging	Offline commands are pending proposals; server re-checks all permissions, versions and business conditions after reconnect.
R7 | Bounded AI	Permission-scoped evidence, deterministic calculations, citations and no privileged write authority.
R8 | Budget controls	No unauthorized commitments beyond approved budget; exception route remains attested and governed.
R9 | Recoverability / tamper evidence	Restore exercises, independent audit anchors, reconciliation and integrity verification.

Open interpretation [DECISION-A01]: whether pre-authorized deterministic journal creation constitutes posting under R3 requires finance/legal/security ADR, updated attestations and targeted regression tests. Until approved, original R3 governs.

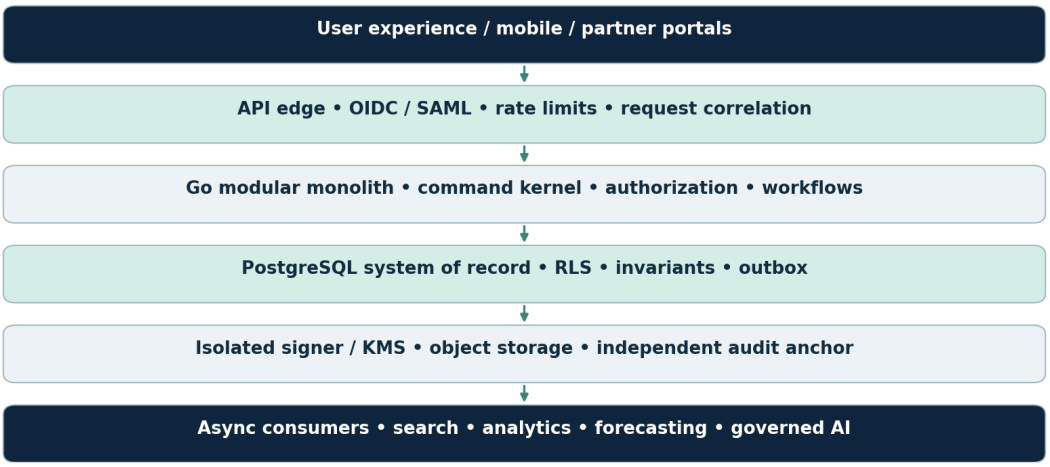
3. Target operating model and role workspaces

Role-based experiences shall expose decision queues and relevant evidence while preventing cross-role access leakage.

Role / workspace	Essential actions	Critical safeguards
CEO / Board	Portfolio exceptions, liquidity exposure, shared-resource conflicts, risk forecasts	Read-only by default; privileged approvals require step-up.
PMO / program director	Cross-project capacity, milestone dependencies, scheduling alternatives	Global reservations authoritative; conflicts cannot disappear via UI.
Project manager	Ready/blocked work, look-ahead, EAC, procurement exceptions	Project-scoped delegation and documented baseline changes.
Site engineer / foreman	Offline diary, quantities, labor, plant, photos, inspection requests	No offline approval/posting; secure queue and conflicts.
QS / commercial	BOQ, change notices, IPC, subcontract valuations, evidence	Effective-dated rates; immutable certified revisions.
Procurement / stores	PR/RFQ/PO, receiving, transfers, stock counts	Budget availability, supplier controls, SoD.
HSE / QA	Permits, ITP, hold points, NCR, incidents, corrective actions	Safety holds can block execution; independent release authority.
CFO / Controller	Close, cash forecast, journal, consolidation, bank reconciliation	Balanced ledger, exact money, period and dual-control rules.
Platform / security	Tenant policies, integrations, role assignment, incident response	No financial authority through administrative role alone.

4. Runtime and deployment architecture

Keep an authoritative Go modular monolith until measured need justifies extraction; prevent distributed coordination from undermining financial invariants.



Container	Authority / failure contract
Browser/mobile/portals	Render, draft, encrypted minimum local cache; never authoritative for balances or permissions.
Identity and edge	OIDC/SAML; device registration; step-up MFA; session lifetime; anti-replay; ingress rate limits; API correlation.
Go domain command kernel	Resolve scope, versions, policies, SoD and risk; validate deterministic domain commands.
PostgreSQL primary	Authoritative aggregates, ledgers, locks, reservation conflict constraints, auditable outbox; RLS defense in depth.
Signer / HSM or KMS	Sign human action + amount + object digest + revision + expiry + policy version; separate security principal.
Object store	Encrypted, scanned, immutable/retained document revisions; controlled signed download links.
Workers and integration	At-least-once delivery; consumer idempotency; bounded retries; DLQ; reconciliation dashboard.
Read models + AI	Async authorized projections with freshness timestamps; failure must not block transactional writes.

4.1 Implementation boundaries

- Domains own their tables, invariants, commands and event schemas. Other domains integrate through application contracts, not ad hoc cross-table writes.
- A command writes aggregate state, ledger effects (if applicable), audit entry and outbox record in one database transaction.

- Reporting may be eventually consistent but every displayed number must identify its source, authorization scope and freshness.
- The production runtime role must not be a table owner, superuser or BYPASSRLS role. Enforce FORCE ROW LEVEL SECURITY where appropriate and test owner/bypass behavior.
- Do not let client-selected tenant, legal entity or project identifiers substitute for server-derived grants.

5. Data model and transaction contracts

The minimum canonical data model should be stable, effective-dated where needed and traceable across all operational and financial events.

Aggregate / entity	Required identifiers and constraints
Organization	tenant_id, legal_entity_id, JV interest/participation, business_unit_id; separate books, tax contexts and permissions.
Project master	project_id, portfolio_id, program_id, contract_id, base_currency, calendar_id, baselines and status.
Work execution	work_package_id, WBS, BOQ_item_id, activity_id, cost_code_id, revision, drawing_id and inspection gate.
Resource reservation	resource_id, window, demand, quantity/capacity, allocation state; overlap exclusion/serializable locking for indivisible units.
Commercial	estimate, approved budget, commitment, approved change, valuation, certification, receivable, retention, claims and forecast separately.
Finance	journal_header_id, line_id, currency, exact amount, entity, period, source_document, reversal_of, approval_attestation.
Documents	document_id, version_id, checksum, access policy, immutable approval state, storage object reference and retention class.
Offline command	client_command_id, device_id, actor, scope, source_version, payload hash, receipt timestamp, reconciliation result.
Domain event	event_id, schema_version, aggregate_id/version, idempotency key, causation/correlation IDs, replay checkpoint.

5.1 Transaction boundary sample

Command: confirm_goods_receipt. Preconditions: valid PO, authorized receiving actor, matching supplier and permitted project, uncontested goods lines, quantity tolerances and approval status. In one DB transaction: confirm GRN version, record inventory movement, update PO received quantity, record required accounting/accrual effects according to policy, append audit proof and outbox event. Retry with same key returns same durable outcome; conflicting payload with same key is rejected.

5.2 API and event principles

Published versioned OpenAPI; problem-detail errors with machine-readable codes; RFC-compatible idempotency key handling; key rotation; resource-based authorization; event contracts with schema compatibility policy. Provide pagination/sorting semantics, bounded export quotas and bulk-processing jobs. Webhooks signed with timestamp, nonce/replay defense and tenant-scoped secrets.

6. Enterprise command, authorization and financial trust

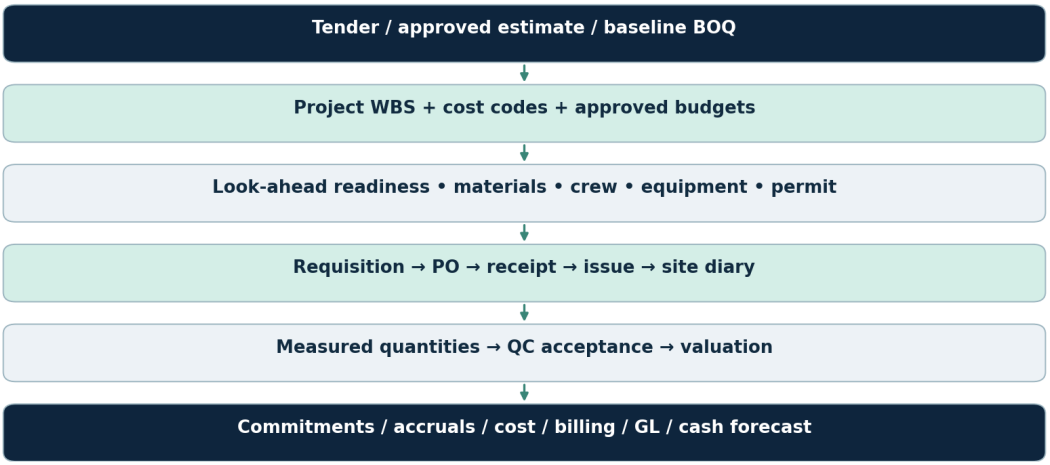
Financial safety is the central engineering constraint; separate user intent, authorization and immutable posting.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
CTRL-01 [P0]	Record and verify exact legal entity, currency, project, source document, role grants and command hash for controlled changes.	Adversarial role-change and replay tests; no unauthorized effect.
CTRL-02 [P0]	Enforce R2/R3 maker/checker and action-bound fresh human attestation for protected decisions.	Self-approval and service-authenticated posting denied at API and SQL boundary.
CTRL-03 [P0]	Balance journal by legal entity/currency as applicable; exact decimal/minor-unit representation, immutable posted state and reversible audit trail.	Property-based accounting tests, double-submit and period-close races.
CTRL-04 [P0]	A committed purchase, stock transfer and resource reservation must not exceed approved allocation without permitted exception.	Concurrent budget and reservation competition accepts only valid totals.
CTRL-05 [P0]	Create external verifiable audit digest anchors independent of operational database administrators.	Privileged rewrite simulation detected during reconciliation.

Caution: use the baseline's exact monetary representation and currency precision rules rather than assuming every currency always has two decimal places. Never conflate "event delivered exactly once" with one effect under retry.

7. Work-package execution engine

The work package is the operational unit connecting what must be built to resources, evidence and commercial value.



State machine: Proposed → Planned → Ready → Released → In Progress → Inspection Hold → Accepted → Completed → Closed. Add Suspended/Cancelled exception states only via governed transitions. Preserve an audit history of every readiness and acceptance change.

Readiness gate	Calculation / block reason	Authority
Approved revision	Current released drawing and correct work-package baseline	Engineering/document controller
Materials	Required item quantities reserved or available within lead-time tolerance	Stores/procurement
Capacity	Qualified crew and plant reserved without conflict	PM/plant manager
HSE / quality	Permits, ITP hold-points and relevant safety approvals valid	HSE/QA approvers
Dependencies	Predecessors complete; access and subcontract conditions satisfied	Planning/project manager
Financial	Budget or permitted variation coverage; no invalid open commitment	Commercial/finance control

Metrics: percent planned complete (PPC), constraint removal lead-time, rework incidence, productivity measured per accepted unit, scheduled vs actual quantity, cost-to-complete and causes of plan failure.

8. Cross-project portfolio optimizer

A portfolio is not a sum of dashboards. It must reason about contested resources and funding while retaining project and legal-entity authority.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
PORT-X01 [P0]	Enforce authoritative global reservation of scarce indivisible plant with collision prevention; cross-entity transfer is policy-controlled.	Two simultaneous projects cannot hold the same excavator in the same time slot.
PORT-X02 [P0]	Model labor skill certification, shift/rest constraints, employment ownership and availability for cross-project assignments.	Unavailable or unqualified crew cannot be confirmed.
PORT-X03 [P0]	Pool purchase demand while preserving source projects, stock ownership, transfer pricing and landed cost allocation.	Source-destination reconciliation exact; no double-issued stock.
PORT-X04 [P1]	Offer three or more explainable what-if allocation choices with cash, schedule and cost trade-offs.	Each scenario lists assumptions, infeasibilities and approval needs.
PORT-X05 [P1]	Render cascading dependencies between critical work packages, projects and programs.	One blocked material delivery surfaces affected milestone chains.

9. Offline-first mobile, connectivity and field trust

The field app must be productive in unreliable connectivity without granting local authority to make high-risk decisions.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
OFF-01 [P0]	Protect minimum local data in encrypted device storage with revocable registration, bounded TTL and authentication.	Lost/revoked device cannot obtain new grants; key lifecycle tested.

OFF-02 [P0]	Durably queue commands with UUID, source version, evidence digest and server-checked actor/project scope.	Offline reboot, duplicate sends and stale revision yield deterministic outcomes.
OFF-03 [P0]	Do not silently merge monetary, stock, certified quantity, permit, safety or approval conflicts.	Explicit conflict queue showing before/after and authorized next action.
OFF-04 [P1]	Sync small commands before large media; resumable attachment uploads and visible freshness.	Constrained 3G simulation, interrupted upload and quota tests.
OFF-05 [P0]	Reject stale/revoked roles and compromised device uploads after reconnection.	Command created before revocation cannot bypass current policy.

10. Detailed domain specifications: commercial and contracts

Commercial management must protect entitlements and evidence throughout the contract lifecycle.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
COM-X01 [P0]	Preserve estimate basis, BOQ rate build-ups, measured quantities, scope change versions and approved baseline history.	Estimate to final account provenance test.
COM-X02 [P0]	Track client contract milestones, notice deadlines, EOT triggers, claims evidence and approval status.	Deadline alerts include source clause and never falsely mark submission.
COM-X03 [P0]	Treat IPC measurement, consultant certification, invoicing, receipt and retention as separate states.	No receivable or revenue recognized solely from an unapproved field diary.
COM-X04 [P1]	Handle subcontract progress, advances, retention, backcharges and variation approvals with SoD.	Reconcile subcontract certification to AP and contract balances.
COM-X05 [P1]	Support project-specific contract forms and clause libraries with legal review.	Contract template change does not retroactively alter executed contracts.

11. Detailed domain specifications: procurement, stock, plant and workforce

Physical asset truth and project cost truth shall be connected without weakening quality or accounting controls.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
RES-X01 [P0]	Maintain PR → RFQ → evaluation → PO → GRN → QC release → inventory issue → invoice match trace.	Source-to-finance reconciliation and negative/duplicate test.
RES-X02 [P0]	Use location, ownership, batch/serial, valuation policy and transfer history for inventory.	Negative stock and double transfers rejected; stock count resolution audited.
RES-X03 [P1]	Record plant assignment, meter hours, fuel, maintenance, breakdowns, lease rates and charge-back rules.	Machine overlap, meter rollback and fuel anomaly detection.
RES-X04 [P1]	Capture certified skills, crew assignment, time, attendance and approved payroll cost distribution.	Project labor totals agree with time and payroll allocations.
RES-X05 [P1]	Support multi-company supply sharing without collapsing distinct ownership or ledgers.	Cross-entity movement creates authorized reciprocal accounting evidence.

12. Detailed domain specifications: planning, progress, quality and HSE

The project schedule and accepted production fact shall be coupled to documented evidence and safety gates.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
DEL-X01 [P0]	Preserve approved schedule baseline, calendar and CPM dependencies with governed replanning.	Critical-path and dependency regression against reference examples.
DEL-X02 [P0]	Measure quantities with unit, method, location, source and verifier; support EVM baseline PV/EV/AC.	Independent recomputation matches reported SPI, CPI and EAC.
DEL-X03 [P0]	Support ITPs, hold points, NCRs, inspections and verified closeout; gate work package release.	Blocked work cannot be closed or certified without required sign-off.

DEL-X04 [P0]	Manage hazard assessments, permits to work, incidents, near misses and corrective actions with escalation.	Critical safety hold cannot be bypassed by PM or AI.
DEL-X05 [P1]	Generate work-complete and handover package with controlled drawings, test results, warranties and as-builts.	A handover export includes provenance and version manifest.

13. Finance, treasury, accounting and consolidated truth

Use the SRS accounting kernel as non-negotiable; add well-defined project commercial subledgers and multi-entity controls.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
FIN-X01 [P0]	Track project budget, approved change, actual, accrual, commitment, forecast and earned progress as distinct dimensions.	No false double-counting or unapproved cost transfer.
FIN-X02 [P0]	AP three-way match and vendor bank-detail modification use independently controlled review.	Account substitution simulation blocks payment release.
FIN-X03 [P0]	Bank payment instructions and bank statement acknowledgment are separate evidence stages.	Unmatched bank transfer cannot be silently settled.
FIN-X04 [P1]	Implement multi-entity journals, intercompany settlement, consolidation and JV accounting via signed policy.	Entity-level books reconcile before consolidated reporting.
FIN-X05 [P0]	Use effective-dated Ethiopian tax and payroll configurations only after adviser sign-off; no hardcoded unverified statutory values.	Legal approval record and golden fixture tests per effective date.

14. Documents, BIM and collaborative project delivery

A CDE is a revision-controlled information environment, not simply a cloud file folder.

ID / class	Mandatory requirement / desired outcome	Acceptance evidence
DOC-X01 [P0]	Version RFIs, submittals, shop drawings, method statements, contracts and transmittals with immutable publication status.	Stale drawing use triggers work-package warning and historical citation.
DOC-X02 [P1]	Provide IFC 4.3 import/reference paths and BCF issues under permission scope, without claiming CAD authoring.	Round-trip loss report and controlled model revision links.
DOC-X03 [P0]	Enforce virus/malware scan, content type validation, preview sandbox, object ACL and retention.	Malicious file and cross-tenant signed URL tests.
DOC-X04 [P1]	External client/supplier/consultant portals have least-privilege project and document access.	Invitation, revocation and scope-drift tests.
DOC-X05 [P1]	Electronic transmittals bind recipient, issue purpose, exact revision, receipt and contractual status.	Disputed version and evidence reconstruction test.

15. AI intelligence and forecasting governance

AI shall explain data and identify possible risks, but calculations and authority must remain deterministic and human-governed.

Class	Output contract	Required evidence
Fact	A previously recorded state	Canonical document/row reference, timestamp, visibility scope
Deterministic calculation	Budget available, CPI, EAC, cash waterfall	Formula version, exact inputs, rounding policy, freshness
Forecast	Potential schedule/cost/stock disruption	Model ID, training cutoff, evaluation, calibration, uncertainty
Recommendation	What-if action proposal	Alternatives, assumptions, approvals needed, side effects

ID / class	Mandatory requirement / desired outcome	Acceptance evidence

AI-X01 [P0]	Permission-filter retrieval before model invocation and recheck before response disclosure.	Prompt-injection and cross-tenant exfiltration suite.
AI-X02 [P0]	No AI tool can authorize cash release, journal posting, change order, safety sign-off or privilege grant.	Attempted tool escalation rejected by kernel.
AI-X03 [P1]	Show evidence, assumptions, confidence/uncertainty and data freshness for forecasts.	Backtest tracked by project type and time period.
AI-X04 [P1]	Use evaluation and rollback gates for new model versions; abstain when evidence unavailable.	Regression suite for hallucination, missing inputs and inappropriate certainty.
AI-X05 [P2]	Support on-prem/air-gapped deterministic analytics with optional local AI, no external data dependency.	Offline deployment operates with AI disabled without breaking core work.

16. Security, privacy and enterprise trust

Threat-model the complete attack surface, not only web/API authentication.

Threat	Defense-in-depth	Evidence needed
Tenant escape	Server-derived tenancy, least-privilege SQL roles, RLS policies, safe exports, scoped search, AI ACL	Cross-tenant attack suite through every data path
Privilege or approval forgery	MFA step-up, fresh signed intent, signer isolation, SoD, replay resistance	API and privileged DB/role compromise simulation
Supplier fraud	Beneficiary controls, out-of-band verification, limit policies and bank evidence	Bank-account substitution tabletop and integration test
Insider audit tampering	Append-only event audit, external digest anchor, controlled retention	Independent consistency verification after privileged rewrite
Malicious documents	Malware scan, MIME checks, preview sandbox and download controls	Oversized/embedded payload corpus
Ransomware / lost region	Isolated immutable backups, credentials isolation, restore and reconciliation	Quarterly timed restore with journal and stock integrity checks

Build compromise	Signed artifacts, dependency scanning, SBOM, SAST, secret scan, deployment provenance	CI evidence and tampered-build rejection
AI injection	Treat retrieval as untrusted, typed tool permissions, evidence gate	Prompt injection and role-exfiltration red team

Technical reference check [VERIFIED-REFERENCE]: OWASP ASVS latest stable edition on 7 Nov 2026 is 5.0.0 (published 2025), replacing the v1 reference to ASVS 4. PostgreSQL documentation confirms owners generally bypass RLS unless FORCE ROW LEVEL SECURITY, and superuser/BYPASSRLS roles bypass. NIST SP 800-207 is an architecture reference, not a certification. See references.

17. Non-functional engineering service levels

Initial targets come from SRS v1, with newly proposed mixed-workload qualification. No capacity or SLO result has been measured by this document.

Target / source	Acceptance / environment
Commands p95 ≤ 1s; p99 ≤ 3s	Measure server latency under v1 reference load with simultaneous writes; account for queueing separately.
List/detail p95 ≤ 500ms; search ≤ 1s	Test role-filtered queries on documented dataset including 5M documents per tenant.
Reports ≤ 10s for ≤ 100k rows	Use async large exports; snapshot timestamp, budgets and ACL.
2,000 concurrent / 500 active projects / 50M journal lines	Production-like sizing, skewed tenants, hot budgets, cross-project reservations, indexes and partition maintenance.
1,000 offline commands / 5 minutes on constrained link	Separate small command payload and large-media sync test methodology; record percentiles and retries.
99.9% SaaS monthly availability	Define SLI and exclusion rules in signed SLA; monitor independently.
RPO ≤ 5 min; RTO ≤ 4h (premium 1h)	Demonstrate restore, financial consistency and actual failure-domain isolation.
WCAG AA / Ethiopic usability	Conformance test and screen-reader/keyboard validation; scripts, calendar and low-connectivity tests.

18. Architecture decisions, technical verification and open evidence

The word verified is only used for official reference confirmation or completed reproducible evidence. Repository code and running deployments were not made available for execution.

Check	What was verified / remains to verify	Gate
Source requirements	All extractable SRS requirement rows preserved verbatim in Appendix A. Baseline counts asserted by source.	Document consistency
New v2 requirements	Structured proposed requirements from v2 retained in Appendix B.	Traceability review
PostgreSQL RLS	Official docs support FORCE RLS owner nuance; actual SERK roles/policies require live test.	P0 security
OWASP ASVS	Official stable release identified as 5.0.0. No SERK ASVS audit conducted.	P0 security
buildingSMART IFC	Official IFC4.3 (4.3.2.0), ISO 16739-1:2024 confirmed; no adapter verification.	P1 interoperability
NIST SP 800-207	Official architecture reference confirmed; no zero-trust audit.	P1 security
Baseline repository	Commit, migrations and 378 tests source-reported, not cloned or rerun.	P0 evidence
Performance and DR	SRS targets only; no load tests, PITR restore or failover execution supplied.	P0 release

19. Acceptance and adversarial scenarios

Every scenario shall identify test fixtures, steps, expected database state, observed logs, reviewer and evidence link.

ID	Test narrative	Acceptance invariant
A01	Two projects reserve the same crane concurrently	One committed reservation, one explained conflict, zero double-booking
A02	Two approvers attempt maker/checker violation	Self-approved or stale attestation attempt rejected
A03	Offline receiving replay after device revocation	Queued request rejected; no stock/ledger mutation
A04	Supplier bank-account substitution before payment	Independent re-verification and release block

A05	Posted journal reversed after close	Original immutable; controlled reversal according to period policy
A06	Cross-tenant search, export, signed media URL and AI request	Zero unauthorized disclosure
A07	Field diary indicates quantity but QA hold persists	No certified progress or payable derived from held work
A08	Budget revision and simultaneous PO award	Atomic coverage check; no unauthorized overcommitment
A09	Database failure after state write before outbox delivery	Transaction entirely committed or rolled back; replay safe
A10	Region outage and restore with multiple projects	Meets proved RPO/RTO; cash/stock/ledger/audit reconcile
A11	Source drawing revision replaced mid-work	Stale work-package readiness invalidated safely
A12	Financial report and AI use stale projection	Freshness shown; source recomputation correct and restricted

20. Release train and implementation program

Release order should de-risk a complete field-to-ledger value chain early without compromising human approval and financial integrity gates.

Release	Mandatory scope	Exit evidence
R0 Control kernel	IAM, RLS, signer, SoD, invariants, audit, backups, CI, migrations	Independent security tests, recovered production-like data, code traceability
R1 Pilot transaction spine	Project, cost codes, BOQ, budget, PR/PO, GRN, stock issue, AP, GL	Single-project end-to-end UAT and accounting reconciliation
R2 Field-to-finance	Offline diary, quantities, work package, inspection holds, progress, valuation	Live field pilot under poor connectivity; zero silent conflicts
R3 Enterprise coordination	Shared plant/crew, multi-project materials, funding forecast, reports, portfolios	Concurrent resource allocation / tenant workload tests
R4 Commercial scale + intelligence	Claims, extensions, BIM, partner portals, forecasts, advanced HSE and QA	Independent performance, privacy, penetration, DR, model risk and operational readiness

Recommended delivery governance: weekly domain design review; ADR owner and immutable decisions; monthly program risk review; quarterly DR/security and field-adoption tests; production change board with CFO and HSE participation for relevant releases. Dates and staffing remain decision-required until an engineering capacity plan exists.

21. Operating metrics and business value

Instrument task outcomes rather than screen counts.

Metric	Definition / target-setting guidance
Work readiness	% weekly packages released with all prerequisites valid; reason-code distribution
Plan reliability	Percent Plan Complete and variance by crew, project, package and weather/access constraints
Commercial capture	% notice deadlines met, variation cycle time, certified-to-invoiced delay
Procurement	PR-to-PO duration, on-time-in-full delivery, emergency purchases, price variance
Cash and cost	Forecast accuracy, receivables aging, project EAC movements, commitment integrity
Field adoption	Median diary completion, offline conflict rate, retry success, forms per role/day
Trust	Unauthorized access attempts blocked, attestation verification failures, RLS test coverage
Reliability	Service SLI, sync freshness, recovery drill pass rate, tenant noisy-neighbor impact

22. Decisions required before baseline approval

The following issues cannot safely be solved by inference alone.

ADR	Decision	Accountable owner	Gate
D01	Risk-tiered authorization vs v1 R2/R3	CFO + Security + Product	Blocking
D02	Legal-entity/JV consolidation treatment and intercompany ownership	Accounting policy owner	Blocking
D03	Currency precision, rounding, WIP and revenue recognition	CFO / qualified accountants	Blocking

D04	Ethiopian tax, labor, privacy, retention and electronic signature applicability	Counsel + tax/payroll advisers	Blocking
D05	Minimum baseline vs portfolio release split and migration sequencing	CTO + PMO	Blocking
D06	Cross-project reservation locks, cancellation priorities and override policy	Operations + CTO	Blocking
D07	Data residency, encryption keys, disaster-recovery region and air-gap constraints	Security + Legal	Blocking
D08	Official CI/CD evidence, code coverage, penetration test and QA signoff	CTO + QA	Blocking
D09	AI evaluation thresholds, allowed models and data processing agreements	AI governance board	Pre-AI launch
D10	IFC/BCF version floor and interoperability validation plan	BIM lead	Pre-integration
D11	UX accessibility and offline hardware minimums	Product/Field lead	Pre-field pilot
D12	Service tier SLOs, cost/capacity and response policy	SRE + Finance	Pre-SaaS launch

23. Definition of done and sign-off

No release may be represented as production certified without current, linked evidence and role-specific approval.

- Functional: requirement implemented, linked to API/schema/UX, reviewed by domain owner, supported by automated and human UAT.
- Security: threat model reviewed, ASVS applicability matrix approved, RLS scope tested across all read/write pathways, independent pen-test findings closed.
- Financial: golden-ledger scenarios and opening-balance/stock reconciliations independently signed by finance.

- Field: online/offline drills under realistic devices/networks, documented conflict support and QA/HSE signoff.
- Operational: alert runbooks, observability, SLI/SLO measurements, restore/PITR evidence, incident exercise and rollback plan.
- Governance: open blocking ADRs closed; signed privacy/tax/accounting decisions; immutable build artifact and deployment approval.

Signatory	Signature / date / status
Product owner	PENDING
Chief Technology Officer	PENDING
Construction operations lead	PENDING
CFO / financial controls owner	PENDING
CISO / security owner	PENDING
Quality and release owner	PENDING
Legal / regulatory adviser as applicable	PENDING

Appendix A. Complete SRS v1.0 baseline requirement register

Copied from source PDF requirement tables, pages 13–30. 253 machine-extracted entries; precise wording, priority, verification type and self-reported baseline status retained except whitespace normalization. Source claims 253 requirements; extraction count checked below. Each row inherits original SRS status and approval process.

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
IAM-01	The system shall support multiple tenants (construction companies). Each tenant's data shall be isolated in application code and by PostgreSQL row-level security, with the tenant derived from the database login and never from a client-settable value.	Must / Test / Exists
IAM-02	The system shall authenticate users through enterprise single sign-on using OpenID Connect and SAML 2.0 (for example Microsoft Entra ID, Google Workspace, Okta, Keycloak). Tenants without an identity provider shall be offered local credentials stored only as Argon2id hashes.	Must / Test / Partial

IAM-03	The system shall support SCIM 2.0 user provisioning and deprovisioning; a deprovisioned user's sessions shall be revoked within 60 seconds.	Should / Test / New
IAM-04	The system shall require multi-factor authentication (WebAuthn/passkeys or TOTP; SMS only if the tenant enables it) for every user holding an approval, posting, payment, security or administration permission, and step-up MFA bound to the specific action within a configurable freshness window (default 10 minutes).	Must / Test / Partial
IAM-05	The system shall provide role-based access control with the seed role catalogue of Appendix D, and shall let tenants compose custom roles from the fixed permission list. A role that would hold both the maker and the checker permission of the same control shall be refused.	Must / Test / Exists
IAM-06	The system shall restrict every read and command to the projects explicitly assigned to the user, unless the user holds a tenant-wide scope permission.	Must / Test / Exists
IAM-07	The system shall evaluate attribute-based rules (legal entity, business unit, project, amount band, document classification, device, principal kind) on every command and read in addition to role checks.	Must / Test / Partial
IAM-08	The system shall enforce segregation of duties (maker differs from checker, and checker differs from poster or releaser where configured) for every controlled document, both in the application and by database constraint.	Must / Test / Exists

IAM-09	The system shall analyse segregation-of-duties conflicts before a role is assigned, against a tenant-maintained conflict rule set, and shall require a documented mitigating control to accept a conflict.	Should / Test / New
IAM-10	The system shall distinguish human, service and AI principals from an identity-provider assertion that no role can grant. Only human principals shall approve, certify, post, pay, release, close or configure.	Must / Test / Exists
IAM-11	The system shall revoke sessions server-side across all API replicas; a role removal shall take effect on the next request.	Must / Test / Exists
IAM-12	The system shall rate-limit authentication and API use with limits shared across replicas, and shall temporarily lock an account after a configurable number of failures and notify the tenant security administrator.	Must / Test / Partial
IAM-13	The system shall maintain a delegation-of-authority matrix (approval limits by role, document type, amount band, entity and project). The matrix shall be versioned and approved by maker/checker.	Must / Test / Partial
IAM-14	The system shall support time-bound, scoped delegation of approval rights (for example during leave). A delegate shall never exceed the delegator's limits, and delegation shall never enable self-approval.	Should / Test / New
IAM-15	The system should provide time-limited break-glass access that requires a typed reason, notifies security administrators and is flagged for review.	Should / Demo / New

IAM-16	The system may support tenant conditional-access policies (IP allow-lists, device posture, country restrictions).	Could / Test / New
ORG-01	The system shall model group, legal entity, region or branch, business unit and project. Each legal entity shall have its own functional currency, tax registration, chart-of-accounts mapping and fiscal calendar.	Must / Test / Partial
ORG-02	The system shall maintain a project master: code, name, client, contract type, location (with GPS coordinates), key dates, project manager, phase, status and stage-gate history.	Must / Test / Partial
ORG-03	The system shall maintain supplier and subcontractor masters with maker/checker approval and blocking. Bank accounts shall be dual-approved with MFA and shown masked unless the user holds the reveal permission (which is audited).	Must / Test / Exists
ORG-04	The system shall maintain a customer master with TIN, credit limit, payment terms and maker/checker approval.	Must / Test / Partial
ORG-05	The system should support supplier and subcontractor pre-qualification: trade licence and contractor grade, tax clearance, VAT registration, insurance and expiry tracking, with alerts before expiry and an evaluated performance rating.	Should / Test / New
ORG-06	The system shall maintain versioned WBS and cost-code structures with maker/checker approval. Documents shall only reference cost codes in an approved version.	Must / Test / Exists

ORG-07	The system shall maintain a resource library (labour trades, plant types, materials, subcontract packages) with units of measure and conversion factors.	Must / Test / Partial
ORG-08	The system should detect probable duplicate partners (same TIN, bank account, or similar name) at creation and route them to review.	Should / Test / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
ORG-09	The system shall maintain working calendars per entity and project, including Ethiopian public holidays, shifts and non-working days.	Must / Test / Partial
EST-01	The system should manage an opportunity and bid pipeline: source, client, value, go/no-go scoring, probability, expected value and bid calendar.	Should / Test / New
EST-02	The system shall manage bills of quantities (BOQ) in versions with items, units, quantities and rates mapped to cost codes. A submitted version shall be immutable; approval shall be by a different person.	Must / Test / Exists
EST-03	The system shall support resource-based estimating: unit-rate build-ups from labour, plant, materials and subcontract resources with productivity factors, plus indirect costs, overhead, risk and margin, drawing on a maintained cost database.	Must / Test / New
EST-04	The system should compare estimate versions and markup scenarios, and price imported resources in their source currency using approved FX rates.	Should / Test / New

EST-05	The system should support quantity take-off from 2D drawings and IFC models with each quantity linked to a BOQ item.	Should / Demo / New
EST-06	The system should manage tender packages: documents, clarifications, addenda, bid-bond tracking and a submission checklist.	Should / Test / New
EST-07	An approved BOQ shall become the project's baseline budget through a controlled transition. It shall never be changed in place.	Must / Test / Exists
EST-08	The system may benchmark unit rates and productivity against completed projects.	Could / Demo / New
PCC-01	The system shall keep the original budget, approved budget revisions, budget transfers between cost codes, and contingency and management-reserve draw-downs, each approved by maker/checker. Original and current budgets shall always be shown side by side.	Must / Test / Partial
PCC-02	The system shall record commitments from approved purchase orders and subcontracts in real time. It shall refuse a requisition or order above the remaining budget of its cost code unless an authorised person overrides it with fresh MFA and a reason.	Must / Test / Exists
PCC-03	The system shall derive actual cost from posted journals by project and cost code. Each figure shall drill down to its source documents.	Must / Test / Exists

PCC-04	The system shall support periodic cost forecasting: estimate to complete and estimate at completion per cost code (manual, CPI-based or remaining-budget methods), forecast versions and an approved monthly forecast cycle.	Must / Test / Partial
PCC-05	The system shall produce a cost report showing, per cost code: original budget, revised budget, committed, actual, accrued, forecast, and variances against budget and the last forecast.	Must / Test / Partial
PCC-06	The system shall produce a work-in-progress schedule per project (percentage complete, earned revenue, billed to date, over- or under-billing) that feeds revenue recognition (FIN-15).	Must / Test / New
PCC-07	The system should manage change events (potential, pending, approved) with cost and schedule impact, linked to client variations and subcontract changes.	Should / Test / New
PCC-08	The system should keep a quantified risk and opportunity register per project and compute contingency at P50 and P80 by Monte Carlo simulation.	Should / Test / New
PCC-09	The system shall roll up project figures by entity, region, business unit, client, contract type and portfolio.	Must / Test / Partial
PCC-10	The system should show project cash flow as planned, actual and forecast S- curves.	Should / Test / New
SCH-01	The system shall compute critical-path schedules with FS, SS, FF and SF relationships, lags, activity calendars and total and free float.	Must / Test / Exists

SCH-02	The system shall keep approved baselines (maker/checker) and compare the current schedule against any of several baselines.	Must / Test / Partial
SCH-03	The system shall support schedule constraints (start no earlier than, finish no later than, mandatory), activity codes and a WBS hierarchy.	Must / Test / New
SCH-04	The system should load resources onto activities, level them, and show labour and plant histograms.	Should / Test / New
SCH-05	The system shall import and export Primavera P6 (XER and XML) and Microsoft Project XML, with an integrity report of anything not carried over.	Must / Test / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
SCH-06	The system should support look-ahead planning (3 to 6 weeks) and weekly work plans with percent-plan-complete tracking.	Should / Test / New
SCH-07	The system should support delay analysis: time-impact fragnets and as-planned versus as-built comparison, linked to extension-of-time claims (CON-04).	Should / Test / New
SCH-08	The system should link schedule activities to model elements for 4D visualisation.	Should / Demo / New
SCH-09	The system may run a schedule quality check (for example the DCMA 14-point assessment).	Could / Test / New
EVM-01	The system shall record progress reports per activity or cost code, approved by a person other than the author.	Must / Test / Exists

EVM-02	The system shall compute PV, EV, AC, SV, CV, SPI, CPI, EAC, ETC and VAC with a documented price basis (nominal or constant) and provenance for every input.	Must / Test / Exists
EVM-03	The system should support rules of credit: weighted milestones, units installed, percent complete, start/finish and level of effort, configurable per cost code.	Should / Test / New
EVM-04	The system should feed installed quantities captured on site (SIT) into progress and client measurement.	Should / Test / New
EVM-05	The system should attach photo and geotag evidence to progress records.	Should / Test / New
PRC-01	The system shall support the flow requisition, RFQ, quotes, technical evaluation, award, purchase order, goods receipt and three-way match, with each step a controlled document.	Must / Test / Exists
PRC-02	The system shall enforce a minimum number of suppliers per RFQ unless there is a recorded single-source justification. An award to other than the lowest compliant quote shall require a reason.	Must / Test / Exists
PRC-03	Purchase-order approval shall follow the delegation-of-authority matrix and budget control (PCC-02).	Must / Test / Exists
PRC-04	The system should support framework agreements and blanket orders with call- offs against agreed price lists.	Should / Test / New
PRC-05	The system should provide e-sourcing through the supplier portal. Quotes shall be sealed until the due time and opened by two people.	Should / Test / New

PRC-06	The system shall manage import procurement: proforma, letter of credit (TRS-05), shipping documents, customs clearance, and allocation of landed cost (freight, insurance, duty, clearing) to received items.	Must / Test / New
PRC-07	The system should track expediting and delivery, and score supplier performance on price, time and quality.	Should / Test / New
PRC-08	The system shall flag duplicate supplier invoices and split purchases just below approval thresholds.	Must / Test / Exists
PRC-09	The system should version purchase-order changes, with re-approval when value, quantity or terms change.	Should / Test / New
SUB-01	The system shall manage subcontract agreements with a schedule of values or rates, retention percentage, advance payment, scope and terms. They shall commit against the budget (PCC-02).	Must / Test / New
SUB-02	The system shall process subcontractor payment applications: measured work, certification by a quantity surveyor other than the preparer, retention, advance recovery, back-charges, withholding tax, and generation of the payable.	Must / Test / New
SUB-03	The system shall manage subcontract variations linked to change events (PCC- 07).	Must / Test / New
SUB-04	The system should track subcontractor compliance (insurance, licences, safety induction) and, if the tenant configures it, block payment while a document is expired.	Should / Test / New

SUB-05	The system should record back-charges and contra-charges with evidence and the subcontractor's acknowledgement.	Should / Test / New
SUB-06	The system should support pay-when-certified rules that link subcontractor payment to client certification.	Should / Test / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
SUB-07	The system should let subcontractors submit applications and documents through the subcontractor portal (COL-03).	Should / Demo / New
INV-01	The system shall manage warehouses and site stores, with receipts, issues to project cost codes, and approved transfers between stores.	Must / Test / Exists
INV-02	The system shall value stock by FIFO or weighted average per tenant policy and reconcile the stock subledger to the inventory control account with zero unexplained difference.	Must / Test / Exists
INV-03	The system shall support stock counts with approval. Adjustments shall generate journals approved and posted by people other than the counter.	Must / Test / Exists
INV-04	The system shall convert units of measure (for example bag to tonne, piece to metre) and store stock in a base unit.	Must / Test / New
INV-05	The system should support material requests and reservations against schedule activities.	Should / Test / New

INV-06	The system should track batches, heat numbers and serial numbers (for example a rebar heat or cement batch) with linked test certificates (QA-04).	Should / Test / New
INV-07	The system should support reorder points, minimum and maximum levels per store, and suggested requisitions.	Should / Test / New
INV-08	The system should support returns to supplier and transfers of surplus between projects at valuation.	Should / Test / New
INV-09	The system should compare material consumption with the theoretical quantity from the BOQ and report wastage.	Should / Test / New
INV-10	The system may print barcode or QR labels and support mobile scanning.	Could / Demo / New
EQP-01	The system shall keep an equipment register (owned, leased, hired) linked to fixed assets (FIN-06), with ownership and operating cost.	Must / Test / New
EQP-02	The system shall charge internal hire rates to projects by recorded usage (hours, days or kilometres) through generated journals.	Must / Test / New
EQP-03	The system shall record daily working, idle and breakdown hours per item of equipment, captured in site diaries or timesheets.	Must / Test / Partial
EQP-04	The system should plan preventive maintenance (by hours, distance or calendar), raise work orders and issue spare parts from inventory.	Should / Test / New
EQP-05	The system should record fuel issues per item of equipment and flag consumption anomalies.	Should / Test / New

EQP-06	The system should record equipment transfers between sites with a condition report at each hand-over.	Should / Test / New
EQP-07	The system may import telematics data (ISO 15143-3 / AEMP 2.0).	Could / Demo / New
HCM-01	The system shall keep an employee master covering salaried staff and daily labourers: contracts, grade, trade, rate, certifications and bank or mobile-money details (masked).	Must / Test / New
HCM-02	The system shall capture timesheets per person or crew against project and cost code, including offline, approved by a supervisor other than the person recording them.	Must / Test / New
HCM-03	The system should capture attendance from biometric devices or a geofenced mobile check-in.	Should / Test / New
HCM-04	The system shall compute payroll with data-driven, effective-dated statutory rules (Ethiopian employment income tax bands, employee and employer pension contributions, overtime under the labour law in force) confirmed by a qualified adviser.	Must / Test / New
HCM-05	Payroll shall be approved by maker/checker and shall generate journals by project and cost code, a bank or mobile-money payment file, and statutory returns.	Must / Test / New
HCM-06	The system should manage leave and overtime requests and approvals.	Should / Test / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status

HCM-07	The system should report labour productivity (units installed per person-hour) against the estimate.	Should / Test / New
HCM-08	The system may support recruitment and onboarding of site workers, with induction records linked to HSE.	Could / Demo / New
CON-01	The system shall manage client contracts with lines or rates, payment terms and a contract policy (retention, advance, VAT base), approved by maker/checker.	Must / Test / Exists
CON-02	The system shall support measurement, interim payment certificate (IPC), receivable invoice and receipt, with retention, retention release and advance recovery computed by the server.	Must / Test / Exists
CON-03	The system shall manage client variations with approval and their effect on contract value.	Must / Test / Exists
CON-04	The system shall keep a claims and extension-of-time register with contractual notices, time bars (for example a 28-day notice period) and alerts before a time bar expires.	Must / Test / New
CON-05	The system shall apply contract price-adjustment formulas based on published indices, per contract.	Must / Test / New
CON-06	The system should compute liquidated damages and bonuses from contract dates.	Should / Test / New
CON-07	The system shall keep a register of guarantees and bonds (bid, advance- payment, performance, retention) with issuer, amount, expiry, reduction and release, and alert before expiry.	Must / Test / New

CON-08	The system should provide configurable contract templates (FIDIC Red, Yellow and Silver Books; Ethiopian public-procurement standard documents) that set default policies.	Should / Test / New
CON-09	The system should keep a contract obligations register with owners and due dates.	Should / Test / New
CON-10	The system should let clients and consultants review measurements and IPCs through the client portal (COL-03).	Should / Demo / New
FIN-01	The system shall record journals as balanced, immutable double entries in whole minor units, with preparer, approver and poster distinct. Corrections shall be by reversal only.	Must / Test / Exists
FIN-02	Each journal generated from a business document shall carry a unique source- event identifier so that no event can be posted twice.	Must / Test / Exists
FIN-03	The system shall manage accounting periods. Posting into a closed period shall be refused, and reopening shall need a privileged, audited permission.	Must / Test / Exists
FIN-04	The system shall support multi-currency transactions with approved rates selected by an approved FX policy, and remeasurement under IAS 21.	Must / Test / Exists
FIN-05	The system shall provide an accounts payable subledger and an accounts receivable subledger with aging, reconciled to their control accounts.	Must / Test / Exists

FIN-06	The system shall manage fixed assets under IAS 16: acquisition, components, depreciation methods, transfers, revaluation, impairment and disposal.	Must / Test / New
FIN-07	The system shall apply data-driven, effective-dated tax rules: output and input VAT, withholding tax on payments and receipts, and excise where applicable. It shall produce the supporting schedules for tax returns.	Must / Test / Partial
FIN-08	The system shall consolidate multiple entities, with intercompany transactions matched and eliminated automatically, and joint ventures accounted for under the tenant's policy.	Must / Test / New
FIN-09	The system shall produce trial balance, general ledger, income statement, balance sheet and cash-flow statement per entity and consolidated, in IFRS presentation.	Must / Test / Partial
FIN-10	The system should support recurring journals, and accruals and prepayments with automatic reversal.	Should / Test / New
FIN-11	The system should allocate overheads to projects by configurable rules.	Should / Test / New
FIN-12	The system should support corporate (overhead) budgets with variance reporting.	Should / Test / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
FIN-13	The system should support the Ethiopian fiscal year (Hamle to Sene) and display dates in the Ethiopian calendar.	Should / Test / New

FIN-14	The system should provide an adapter for tax-authority e-invoicing and fiscal- receipt obligations, once their applicability and format are confirmed (Appendix E).	Should / Analysis / New
FIN-15	The system shall recognise contract revenue over time under IFRS 15 (input or output method per contract) and maintain contract assets and liabilities.	Must / Test / New
TRS-01	Payments shall be prepared, approved (dual approval above a threshold) and released by different human principals with fresh MFA. The system shall never move money itself; settlement shall be recognised only from imported bank statements.	Must / Test / Exists
TRS-02	The system shall forecast cash for 13 weeks and for the project horizon, separating confirmed from forecast flows and including payroll, taxes, supplier schedules, expected IPC receipts and debt service.	Must / Test / Partial
TRS-03	The system should exchange payment files (ISO 20022 pain.001) and statements (camt.053, MT940 or bank CSV) with banks where they support it.	Should / Test / New
TRS-04	The system should track bank facilities, overdrafts and guarantee lines, and their utilisation.	Should / Test / New
TRS-05	The system should manage letters of credit for imports, from application to settlement.	Should / Test / New
TRS-06	The system may manage site petty cash on an imprest basis, with replenishment approvals.	Could / Test / New

SIT-01	The system shall record daily site diaries (weather, labour, equipment, notes, instructions), approved by a person other than the author.	Must / Test / Exists
SIT-02	Only registered, revocable devices shall sync offline work. Sync shall be idempotent per device and operation id, and a conflict shall be returned to the user, never merged silently.	Must / Test / Exists
SIT-03	The system shall provide an offline-first mobile application (Android first, then iOS) for diaries, timesheets, goods receipts, inspections, incidents, photos and approvals.	Must / Demo / New
SIT-04	Offline devices shall prepare commands only. They shall never create authoritative financial state.	Must / Test / Exists
SIT-05	The system should capture photos with GPS position, time and watermark, and link them to any record.	Should / Test / New
SIT-06	The system should manage punch and snag lists with assignees, due dates and close-out photos.	Should / Test / New
SIT-07	The system should record site instructions and meeting minutes with tracked actions.	Should / Test / New
SIT-08	The system may link drone or 360-degree captures to locations and dates.	Could / Demo / New
DOC-01	The system shall provide a common data environment following ISO 19650 (work in progress, shared, published, archived), with tenant-defined numbering rules and metadata.	Must / Test / New
DOC-02	The system shall manage document revisions. A superseded revision shall never be presented as current.	Must / Test / New

DOC-03	The system shall issue transmittals and record each recipient's acknowledgement.	Must / Test / New
DOC-04	The system shall manage RFIs, with ball-in-court, due dates, responses, and links to cost and schedule impact.	Must / Test / New
DOC-05	The system should manage submittals (shop drawings, material approvals) with review status and stamps.	Should / Test / New
DOC-06	The system should keep a drawing register and allow mark-up in the browser.	Should / Demo / New
DOC-07	The system should view IFC models in the browser and exchange model-based issues in BCF.	Should / Demo / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
DOC-08	Files shall be encrypted at rest, scanned for malware and served only through short-lived signed URLs after an authorisation check.	Must / Test / New
DOC-09	The system should provide permission-filtered full-text search over documents, including text extracted by OCR.	Should / Test / New
QA-01	The system shall manage inspection and test plans per activity or work package, with hold and witness points.	Must / Test / New
QA-02	The system shall record inspection requests and results. If the tenant configures it, progress or measurement on an activity with an unreleased hold point shall be refused.	Must / Test / New
QA-03	The system shall manage non-conformance reports with root cause, corrective and preventive action, cost impact and verified closure.	Must / Test / New

QA-04	The system should record material test results (for example concrete cubes or rebar tensile tests) linked to batches (INV-06).	Should / Test / New
QA-05	The system should feed quality outcomes into supplier and subcontractor ratings.	Should / Test / New
HSE-01	The system shall record incidents and near misses (including offline), with investigation, root cause and corrective actions tracked to closure.	Must / Test / New
HSE-02	The system shall manage permits to work (hot work, confined space, work at height, excavation, lifting) with approvals, conditions and expiry.	Must / Test / New
HSE-03	The system should record inductions, toolbox talks and training, and can be configured to block site access or timesheets for workers who have not been inducted.	Should / Test / New
HSE-04	The system should run safety inspection and observation checklists.	Should / Test / New
HSE-05	The system shall compute HSE indicators (LTIFR, TRIR, severity rate) using hours worked from timesheets.	Must / Test / New
HSE-06	The system may record environmental aspects: waste, spills and environmental permits.	Could / Test / New
WFL-01	The system shall execute versioned workflow definitions declaring steps, permissions, quorums, SoD rules, MFA, amount thresholds, SLA escalation and expiry. A validator shall refuse any definition that weakens a control.	Must / Test / Exists

WFL-02	When a definition is published as mandatory, the database shall refuse the subject's sensitive transition unless the same transaction recorded the workflow effect with its full quorum.	Must / Test / Exists
WFL-03	Every human decision shall carry an attestation signed by the isolated approval signer and verified by the database.	Must / Test / Exists
WFL-04	The system shall present one task inbox across all modules, on web and mobile, with step-up MFA for sensitive decisions.	Must / Test / Partial
WFL-05	The system should provide a visual workflow designer that produces the same validated definition format.	Should / Test / New
WFL-06	Automation rules shall only notify or prepare drafts. The rule's author and its activator shall be different people.	Must / Test / Exists
WFL-07	The system should redirect tasks on absence through approved delegations (IAM-14).	Should / Test / New
RPT-01	The system shall provide a maintained catalogue of standard reports across all modules (at least 60 at general availability; currently 11 of the 25 initially listed exist).	Must / Test / Partial
RPT-02	Every report and list shall export to PDF, Word and Excel with the tenant's letterhead. Exports shall contain no more than the user can see, and shall be watermarked and audited.	Must / Test / Exists
RPT-03	The system shall provide role dashboards (executive, project, finance, procurement, HSE). Every KPI shall have a documented formula and drill down to its source documents.	Must / Test / Partial

RPT-04	The system should offer a self-service report builder over a governed semantic layer, with saved filters and views.	Should / Test / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
RPT-05	The system should deliver reports by email on a schedule, subject to the recipient's permissions.	Should / Test / New
RPT-06	Every report shall display its filters and data-freshness timestamp.	Must / Test / Partial
RPT-07	The system should provide a read-only analytics feed (warehouse export and BI connectors such as Power BI) that preserves tenant and project isolation.	Should / Test / New
RPT-08	The system should produce statutory reports: VAT return, withholding tax, pension and payroll tax schedules.	Should / Test / New
AI-01	The system should answer operational questions (cost overrun, schedule slip, cash, overdue orders) from permission-filtered facts, citing sources, freshness, assumptions and missing data.	Should / Demo / Partial
AI-02	AI principals shall not approve, certify, post, pay or release. Their only write capability shall be preparing drafts for a human.	Must / Test / Exists
AI-03	All numbers shown shall come from deterministic computation. Any language-model phrasing shall be checked against the facts and discarded if it introduces a number not in them.	Must / Test / Exists

AI-04	The system should extract data from supplier invoices, delivery notes and certificates into draft documents with per-field confidence. A person shall confirm each draft.	Should / Test / New
AI-05	The system should predict cost-overrun, schedule-slip, cash-shortfall and supplier-delay risk with explanations, and record each prediction so it can be compared with the rule-based baseline and with outcomes.	Should / Test / Partial
AI-06	Untrusted content (documents, OCR output, free text) shall be treated as data, never as instructions. Prompt-injection red-team tests shall run in CI.	Must / Test / Partial
AI-07	Use of an external model shall require tenant opt-in, data minimisation and an approved data-residency decision. A tenant administrator shall be able to disable all AI features.	Must / Inspection / New
AI-08	AI interactions (user, question, facts used, timestamp) shall be logged without unnecessary personal data.	Must / Inspection / Partial
AUD-01	The system shall write a hash-chained, append-only audit record for every command, decision, export, sensitive-data reveal, login and configuration change.	Must / Test / Exists
AUD-02	Every business change shall write its audit record and outbox event in the same transaction.	Must / Test / Exists
AUD-03	Posted records shall be immutable to every application login and to the schema owner (database triggers). An integrity scan shall detect out-of-band tampering.	Must / Test / Exists

AUD-04	Audit logs shall be readable only by authorised roles, and reading them shall itself be audited. External auditors shall get time-boxed, read-only access.	Must / Test / Partial
AUD-05	The system shall provide a control review report: SoD exceptions, budget overrides, approvals outside SLA, reopened periods and revealed bank details.	Must / Test / Partial
AUD-06	The system shall trace end to end from payment to invoice, receipt, purchase order, award, requisition, budget line and BOQ item, and from receipt to IPC, measurement and contract line.	Must / Test / Partial
AUD-07	The system should generate a period evidence pack for the external auditor.	Should / Analysis / New
COL-01	The system shall raise in-app notifications from outbox events after commit.	Must / Test / Exists
COL-02	The system should deliver notifications by email, SMS and messaging channels (for example Telegram), configurable by role and user preference, with retry.	Should / Test / New
COL-03	The system should provide external portals for clients, consultants, suppliers and subcontractors. Portal identities shall be limited to their own records and actions.	Should / Test / New
COL-04	The system should support comments and mentions on any record, retained with the record's audit trail.	Should / Test / New
INT-01	All APIs shall be versioned, documented in OpenAPI 3, authenticated and rate- limited. Every command shall accept an idempotency key.	Must / Inspection / Exists

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
INT-02	Domain events shall be delivered as signed webhooks from the transactional outbox, with per-aggregate ordering, retries, a dead-letter queue and replay.	Must / Test / Exists
INT-03	The system should offer bulk import and export (CSV and Excel templates) through a validated staging area with review before commit.	Should / Test / New
INT-04	The system should provide connectors for Primavera P6, Microsoft Project, Microsoft 365 and SharePoint, Google Drive and Power BI.	Should / Test / New
INT-05	The system should connect to Ethiopian commercial banks and to mobile-money services (for example telebirr) where they offer interfaces.	Should / Test / New
INT-06	External adapters shall be isolated behind timeouts, retries with back-off and circuit breakers, so that no outside failure blocks internal transactions.	Must / Test / Partial
INT-07	The system should provide migration tooling from common legacy systems (Peachtree/Sage, QuickBooks, Excel, SAP Business One), with reconciliation reports.	Should / Test / New
ADM-01	The system shall provide guided tenant onboarding: entities, chart of accounts (construction IFRS template), taxes, banks, users, roles and policies.	Must / Demo / Partial
ADM-02	Operational policies (P2P, contract, stock, FX, thresholds) shall be versioned, maker/checker approved and audited.	Must / Test / Exists

ADM-03	The system shall manage company settings: letterhead, document numbering, locale and portal branding.	Must / Test / Partial
ADM-04	The interface shall be available in English and Amharic at launch. Afaan Oromo and Tigrinya shall be addable without code changes. Translation shall never alter canonical data.	Must / Test / Partial
ADM-05	The system should let administrators add custom fields per document type, which shall be validated and available in reports.	Should / Test / New
ADM-06	The system should show administrators backup, queue, sync and integration health.	Should / Demo / Partial
ADM-07	The system should create sandbox copies of a tenant, with personal data masked, for training and testing.	Should / Test / New
ADM-08	The system shall support configurable retention periods and legal holds per record type.	Must / Test / New
NFR-PERF-01	Command endpoints shall respond within 1 s at the 95th percentile and 3 s at the 99th percentile under the reference load.	Must / Test / New
NFR-PERF-02	Lists and detail reads shall respond within 500 ms at the 95th percentile. Search shall respond within 1 s over 5 million documents per tenant.	Must / Test / New
NFR-PERF-03	Reports over up to 100,000 rows shall complete within 10 s. Larger reports shall run asynchronously and notify the user.	Should / Test / New
NFR-PERF-04	The reference deployment shall support 2,000 concurrent users, 500 active projects per tenant and 50 million journal lines per tenant without breaching NFR-PERF-01.	Must / Test / New

NFR-PERF-05	A device with 1,000 queued offline commands shall finish syncing within 5 minutes on a 3G-class connection.	Should / Test / New
NFR-PERF-06	Over a constrained connection, the web console shall load in under 1.5 MB (compressed) on first visit.	Should / Test / New
NFR-AVL-01	The SaaS service shall achieve 99.9 % monthly availability for transactional services, excluding announced maintenance (99.5 % during the first six months after launch).	Must / Analysis / New
NFR-AVL-02	The recovery point objective shall be at most 5 minutes (point-in-time recovery) and the recovery time objective at most 4 hours, or 1 hour for the premium tier.	Must / Test / Partial
NFR-AVL-03	An interruption at any point shall leave each transaction fully committed or fully rolled back. Replaying the request shall be safe.	Must / Test / Exists
NFR-AVL-04	A failure of reporting, AI, notification, export or integration services shall not prevent transactions.	Must / Test / Partial

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
NFR-AVL-05	Backups shall be encrypted, copied to an isolated off-site location in object-locked storage, and restore-tested at least quarterly, with a signed record of each test.	Must / Test / Partial
NFR-AVL-06	Deployments and migrations shall be backward-compatible so that releases need no downtime during working hours.	Should / Inspection / New

NFR-SEC-01	All traffic shall use TLS 1.2 or higher (TLS 1.3 preferred), with HSTS on public endpoints.	Must / Inspection / Partial
NFR-SEC-02	Data at rest shall be encrypted. Bank details and personal identifiers shall also have field-level encryption.	Must / Inspection / Partial
NFR-SEC-03	The system shall meet OWASP ASVS Level 2, with Level 3 for authentication, approvals, payments and administration.	Must / Analysis / Partial
NFR-SEC-04	The approval-signing key shall be held in an HSM or cloud KMS under a documented key ceremony. Other secrets shall be held in a vault and rotated. No secret shall be in source control or client code.	Must / Inspection / Partial
NFR-SEC-05	The CI pipeline shall run SAST, dependency vulnerability scanning (govulncheck), secret scanning, checksum verification and SBOM generation, and shall sign container images (target SLSA level 3).	Must / Inspection / Partial
NFR-SEC-06	Automated cross-tenant and privilege-escalation probes shall run in CI against the database as a tenant login and as the schema owner.	Must / Test / Exists
NFR-SEC-07	An independent penetration test shall close all critical and high findings before general availability, and shall be repeated at least annually.	Must / Analysis / New
NFR-SEC-08	Security events shall be logged centrally (SIEM export) with alerts. Logs shall never contain passwords, tokens or bank details.	Must / Inspection / Partial
NFR-SEC-09	Exports shall be subject to per-user volume budgets, watermarks and anomaly alerts (anti-exfiltration).	Must / Test / Partial

NFR-SEC-10	The vendor should obtain ISO/IEC 27001 certification and a SOC 2 Type II report within 18 months of general availability.	Should / Analysis / New
NFR-PRV-01	The system shall support compliance with the Ethiopian Personal Data Protection Proclamation No. 1321/2024 (minimisation, purpose limitation, data-subject rights, breach notification, cross-border restrictions), and with GDPR for tenants that need it.	Must / Analysis / New
NFR-PRV-02	Tenants shall be able to choose in-country hosting or on-premises deployment (data residency).	Must / Analysis / New
NFR-PRV-03	Retention periods for financial, payroll, contract and HSE records shall be set to legal minimums confirmed by counsel. Deletion shall be by approved purge or anonymisation only.	Must / Analysis / New
NFR-PRV-04	A documented incident and breach-response procedure, and a data-processing agreement template, shall exist before general availability.	Must / Inspection / New
NFR-PRV-05	Financial and tax treatment (IFRS, VAT, withholding, payroll) shall be reviewed and signed by a qualified accountant or tax adviser before live use.	Must / Analysis / Partial
NFR-USB-01	Non-mobile screens shall conform to WCAG 2.1 level AA, and every function shall be usable by keyboard.	Must / Inspection / Partial
NFR-USB-02	A clerk shall be able to work a standard document (requisition, goods receipt, invoice) after no more than one day of training, to be validated in UAT.	Should / Demo / New

NFR-USB-03	Layouts shall adapt from desktop to tablet and phone. Approvals and dashboards shall be fully usable on a phone.	Must / Test / Partial
NFR-USB-04	Ethiopic script shall render correctly on screen and in every export format.	Must / Test / Partial
NFR-USB-05	Users should be able to display dates in the Ethiopian or Gregorian calendar. Timestamps shall be stored in UTC and displayed by default in the Africa/Addis_Ababa time zone.	Should / Test / New
NFR-USB-06	The server's refusals shall be explained in plain language that names the rule and the action that would satisfy it.	Must / Test / Exists
NFR-USB-07	The product should apply one documented design system across web, mobile and portals.	Should / Inspection / New

Source ID	Exact baseline requirement (whitespace normalized)	Pri. / Verify / Status
NFR-MNT-01	The system shall be a modular monolith with explicit module boundaries. Modules shall not write each other's tables directly.	Must / Inspection / Exists
NFR-MNT-02	Database migrations shall be additive and checksum-locked. A committed migration shall never be edited.	Must / Inspection / Exists
NFR-MNT-03	Automated tests shall cover every invariant R1 to R9, and line coverage of the financial, approval, stock, contract and sync modules shall be at least 80 %.	Must / Test / Partial
NFR-MNT-04	The system shall be delivered with an OpenAPI specification, ER diagram, ADRs and operational runbooks, kept current with each release.	Must / Inspection / Partial

NFR-MNT-05	The system shall be containerised and deployable as multi-tenant SaaS, as a dedicated single-tenant instance, and on premises (including air-gapped), depending only on PostgreSQL and S3-compatible storage.	Must / Demo / Partial
NFR-MNT-06	The system shall emit structured logs, metrics and OpenTelemetry traces with correlation IDs, with SLO-based alerts for availability, latency, outbox lag and ledger integrity.	Must / Inspection / Partial
NFR-MNT-07	API replicas and workers shall scale horizontally. Large tenants shall be placeable on a dedicated database.	Should / Test / New
NFR-MNT-08	The supported clients shall be the current and previous major versions of Chrome, Edge, Firefox and Safari, and Android 10 or later for the mobile application.	Must / Test / Partial
NFR-DAT-01	Critical invariants shall be enforced by database constraints and triggers: balance, uniqueness of source events, immutability, period state, tenant identity and non-negative stock.	Must / Inspection / Exists
NFR-DAT-02	Financial, contractual and audit records shall never be hard-deleted.	Must / Test / Exists
NFR-DAT-03	Money shall use exact decimal types (never floating point), with explicit and tested rounding and allocation.	Must / Test / Exists
NFR-DAT-04	The server clock shall be authoritative. Device-claimed times shall be stored as evidence beside server receipt times.	Must / Test / Exists

NFR-DAT-05	A legacy data migration shall reconcile opening balances, open documents and stock to the source system's totals before go-live.	Should / Test / Partial

Appendix B. v2.0 proposed requirement register

41 structured proposal rows extracted from original v2.0 architecture document. Each remains proposed until mapped, approved and tested; some may overlap with new design requirements in main body.

ID	Priority	Proposed behavior	Proposed verification
PORT-01	P0	Enterprise hierarchy: Model tenant, legal entity, JV, portfolio, program and project with scoped authorization.	Create and query 100 projects in 3 entities; denied scope never leaks via search/export.
PORT-02	P0	Plant allocation: Prevent concurrent confirmed reservations of the same indivisible resource.	Two competing requests cannot both commit under load; conflict explanation logged.
PORT-03	P0	Labor allocation: Support crew skills, availability, certifications, project assignments and conflicts.	Scheduling reports detect overlapping shifts and certification expiry.
PORT-04	P0	Inventory pooling: Allow cross-project stock reservation and controlled value-preserving transfers.	Source and destination inventory and project accounts reconcile after transfer.
PORT-05	P0	Funding waterfall: Compute group cash timing and allocation conflicts with currency/source timestamps.	Drill from consolidated exposure to authorized project cash inputs.

PORT-06	P1	Scenario planning: Compare at least 3 allocation alternatives with explicit assumptions and constraints.	User can see feasibility violations and quantifiable portfolio impacts.
PORT-07	P1	Project templates: Clone governed WBS/BOQ/workflow roles/schedule templates with scope-specific overrides.	Changes never mutate previously baselined projects.
PORT-08	P1	Dependency graph: Maintain cross-project shared-resource and program-milestone dependencies.	Risk to one item identifies affected downstream packages and projects.
EXEC-01	P0	Work readiness: Compute work-package readiness from materials, drawings, skills, plant, permits and holds.	Blocking reason has source, owner and authorized corrective path.
EXEC-02	P0	Site to finance: Use accepted site facts to support progress, valuation and cost without redundant re-entry.	Accepted quantities trace to source device and supporting evidence.
EXEC-03	P0	Field continuity: Encrypted offline drafts with durable queue, device revocation and conflict handling.	Recover from 3G interruptions, duplicate commands and stale versions without double effects.
EXEC-04	P1	Look-ahead: Plan and measure weekly production commitments against readiness constraints.	PPC and failure causes linked to activities/work packages.
EXEC-05	P1	Sector templates: Add building, roads, water, MEP and industrial templates via configuration.	Construction-specific workflows do not fork the core posting logic.

COM-01	P0	Claims risk: Track notice deadlines, notices sent, evidence and claim status.	Missed notices surfaced before deadline; no false auto-submission.
COM-02	P0	Change control: Separate potential, pending, authorized and recognized contract changes.	No unapproved variation is counted as certified revenue.
COM-03	P1	Productivity: Compute units installed per crew-hour/resource baseline and explain variance.	Source metric trace and denominator rules verified on fixture data.
UX-01	P0	Role workspaces: Provide My Work, approvals, project switcher, global command and favorites.	Target task completion validated by role-based UAT.
UX-02	P0	Local usability: Provide local language rendering, Ethiopian date display and compressed site journeys.	Language and offline accessibility test fixtures pass.
UX-03	P1	Voice capture: Speech-to-draft notes with user confirmation and clear AI disclosure.	No factual record auto-commits before verification.
UX-04	P1	Guided onboarding: Configure organization, roles, sample project, source imports and approvals through a wizard.	Pilot tenant setup checklist passes without database manual edits.
DATA-01	P0	Canonical keys: Adopt stable entity, project, cost, WBS, BOQ, activity and document revision IDs.	End-to-end source lineage from cost report to input document.
DATA-02	P0	Read freshness: Show freshness and source version for analytics, forecasts and AI results.	Lagging projection never masquerades as live authoritative balance.

DATA-03	P0	Concurrency: Protect shared budget and allocation invariants from simultaneous commands.	Race tests preserve uniqueness and budget limits.
DATA-04	P1	Event governance: Version external event schemas and maintain replay/consumer deduplication.	Duplicate/out-of-order event tests yield one business effect.

ID	Priority	Proposed behavior	Proposed verification
SEC-11	P0	Audit anchoring: Anchor signed digest chain beyond a compromised DB administrator boundary.	Tampered local records detected against independently held digest.
SEC-12	P0	Device security: Implement encryption, device revocation, minimum offline cache and wipe policy.	Expired/revoked session cannot commit stale commands.
SEC-13	P0	Transaction-bound MFA: Bind approvals to exact action, amount, entity and document revision.	Attestation fails if content or approval policy changes.
SEC-14	P0	Supplier payment fraud: Require verified bank changes and monitor high-risk beneficiary modification.	Fraud-scenario acceptance testing and independent approvals.
SEC-15	P1	Trust boundary testing: Test all data stores, files, exports and AI retrieval for cross-tenant leaks.	CI red-team suite includes schema owner, device and model paths.
AI-09	P0	Response labeling: Distinguish evidence facts, calculations, predictions and recommendations.	Every response links source, assumptions and freshness; predictions labeled.

AI-10	P0	Model permission: Models only access permission-filtered read/draft tools under explicit policy.	Prompt injection cannot approve, pay, alter ACL or read hidden projects.
AI-11	P1	Model evaluation: Record measurable predictive accuracy, false alarms and human outcomes.	Holdout test and drift alerts established before operational reliance.
AI-12	P1	Document intake: Extract source-backed invoice, certificate and delivery drafts with review UI.	Confidence per field, provenance and rejection correction captured.
OPS-09	P0	Portfolio load: Certify mixed load with shared-resource contention and data growth.	Meet published latency/availability SLO with 500 projects target.
OPS-10	P0	Recoverability: Test immutable isolated restore including audit chain and project finance checks.	Signed restore and reconciliation evidence at go-live.
OPS-11	P1	Integration ops: Expose health, replay, dead letters, field mapping and reconciliation console.	Operator can safely repair stalled integration with audit trail.
GOV-01	P0	Policy inheritance: Support governed tenant→entity→project policy inheritance and approved exceptions.	Project cannot silently weaken a mandatory corporate control.
GOV-02	P0	Release evidence: Link each P0 requirement to tests, pilot owner, threat model and sign-off.	No mandatory unresolved gate or critical defect at GA.
GOV-03	P1	Decision control: Track ADR owners for mobile stack, residency, finance, contract and AI.	Every blocking open decision approved before release freeze.

ESG-01	P2	Waste tracking: Measure planned, consumed and wasted materials with location and disposal record.	Quantities reconcile to stock and project reports.
ESG-02	P2	Handover assets: Provide digital asset/equipment register and verified commissioning package.	Client handover references approved model/document versions.